

OneEmo: A Unified Multimodal Reasoning Model for Emotion Perception, Understanding, and Interaction

Jiahao Huang, Zheng Lian *Senior Member, IEEE*, Jingyi Zhang, Zhide Chen, Xiaojiang Peng, Shaonan Wang

Abstract—Multimodal Large Language Models (MLLMs) have demonstrated remarkable capabilities in emotional intelligence. However, prevailing research predominantly focuses on task-specific specialization, often neglecting inter-task synergy and leaving latent reasoning potential underexplored. To bridge this gap, we introduce *OneEmo*, a unified affective generalist capable of mastering emotion perception, comprehension, and interaction. For this purpose, we first construct *EmoWorld-130K*, a comprehensive dataset that distills specialized affective knowledge into explicit reasoning trajectories via a human-in-the-loop workflow. Supervised fine-tuning on this corpus reveals significant mutual benefits derived from multi-task learning. Second, to fully unlock the latent reasoning potential, we propose *Emo-Chord*, a novel reinforcement learning strategy that stabilizes optimization through unified multi-task reward allocation. Extensive experiments demonstrate that *OneEmo* achieves state-of-the-art performance against similarly sized baselines across most benchmarks. Notably, despite having significantly fewer parameters than commercial models, *OneEmo* delivers highly competitive results. This paper paves the way for more reliable and interpretable affective computing. The code is available at <https://github.com/waHAHJIAHAO/OneEmo>.

Index Terms—Affective computing, *EmoWorld-130K* (theory-driven reasoning dataset), *Emo-Chord* (multi-task training strategy), *OneEmo* (unified emotion reasoning model).

I. INTRODUCTION

AFFECTIVE computing has demonstrated significant impact across diverse domains, including educational assistive systems [1], healthcare [2], and conversational agents [3]. At the forefront of this field, multimodal large language models (MLLMs) have substantially advanced the development of affective computing. Recent studies have investigated the application of MLLMs to a spectrum of emotion-related tasks, ranging from foundational emotion perception and understanding [4], [5] to advanced interaction-level competencies, such as the generation of human-like empathy [6], [7].

Early efforts have predominantly relied on supervised fine-tuning (SFT). Specifically, these approaches curate instruction-following datasets that map video inputs to emotion labels,

followed by additional SFT on pre-trained MLLMs to bolster emotion understanding. Representative frameworks include AffectGPT [5] and Emotion-LLaMA [4], which prioritize multimodal fusion and emotion-specific multimodal inputs, respectively. Despite their efficacy, supervised paradigms are inherently limited in eliciting the latent reasoning potential of MLLMs. To bridge this gap, inspired by the remarkable success of reinforcement learning (RL) in reasoning tasks [8], researchers are increasingly integrating RL paradigms into affective computing. Such endeavors have yielded consistent performance gains, enhancing MLLM capabilities in basic emotion recognition [9]–[12], fine-grained emotion perception [13], and multi-turn affective dialogue [14].

However, existing approaches predominantly focus on task-specific emotion specialists, often overlooking inter-task synergy. In human-computer interaction (HCI), emotion perception, understanding, and the execution of appropriate emotion interactions are seamlessly integrated [15]. Given their strong correlations, recent efforts such as Nano-EmoX [12] and VidEmo [16] have begun to leverage this synergy. *Nevertheless, current research still falls short of treating perception, understanding, and interaction as a joint optimization objective, leaving the development of a unified emotion foundation model underexplored.* Furthermore, adapting RL paradigms to such a holistic framework presents significant challenges. Current emotion reasoning datasets are largely confined to perception tasks, lacking explicit reasoning trajectories for higher-order cognitive processes (e.g., intent inference and dialogue generation). Meanwhile, integrating heterogeneous tasks via RL introduces severe optimization hurdles. Optimizing under disparate reward signals frequently triggers gradient conflicts and policy collapse, thereby destabilizing convergence and hindering the realization of a robust affective generalist.

To address these challenges, we propose **OneEmo**, a unified reasoning model explicitly designed for emotional intelligence. To achieve this, we introduce **EmoWorld-130K**, a multi-level reasoning dataset spanning from emotion perception to interaction, and **Emo-Chord**, a novel multi-task reinforcement learning strategy that mitigates model collapse. As illustrated in Fig. 1, *EmoWorld-130K* encompasses eight core affective tasks: Multimodal Sentiment Analysis (MSA), Basic Emotion Recognition (B-MER), Open-Vocabulary MER (OV-MER), Intent Recognition (MIR), Humor Understanding (MHU), Sarcasm Understanding (MSU), Empathetic Response Generation (ERG), and Emotional Support Conversation (ESC). Each instance is annotated with theory-driven, structured reasoning trajectories and corresponding answers. To overcome multi-task optimization instability, we propose *Emo-Chord*, a novel RL strategy that employs an off-policy cold start followed by

Jiahao Huang is with the Fujian Normal University, Fujian, and the State Key Laboratory of Autonomous Intelligent Unmanned Systems, Tongji University, Shanghai (e-mail: qsz20241923@student.fjnu.edu.cn). Zheng Lian is with the State Key Laboratory of Autonomous Intelligent Unmanned Systems, Tongji University, and the Frontiers Science Center for Intelligent Autonomous Systems, Ministry of Education (e-mail: lianzheng@tongji.edu.cn). Jingyi Zhang is with the School of Psychology, Fujian Normal University (e-mail: kr1st3n0104@gmail.com). Zhide Chen is with the Provincial Key Lab of Network Security & Cryptology, Fujian Normal University, Fujian (e-mail: zhidechen@fjnu.edu.cn). Xiaojiang Peng is with Shenzhen Technology University, China (e-mail: pengxiaojiang@sztu.edu.cn). Shaonan Wang is with the Division of Artificial Intelligence and the Humanities, and the Department of Language Science and Technology, Hong Kong Polytechnic University, China (e-mail: shaonan.wang@polyu.edu.hk). Project Leader: Zheng Lian. Corresponding authors: Zheng Lian, Zhide Chen, and Shaonan Wang.

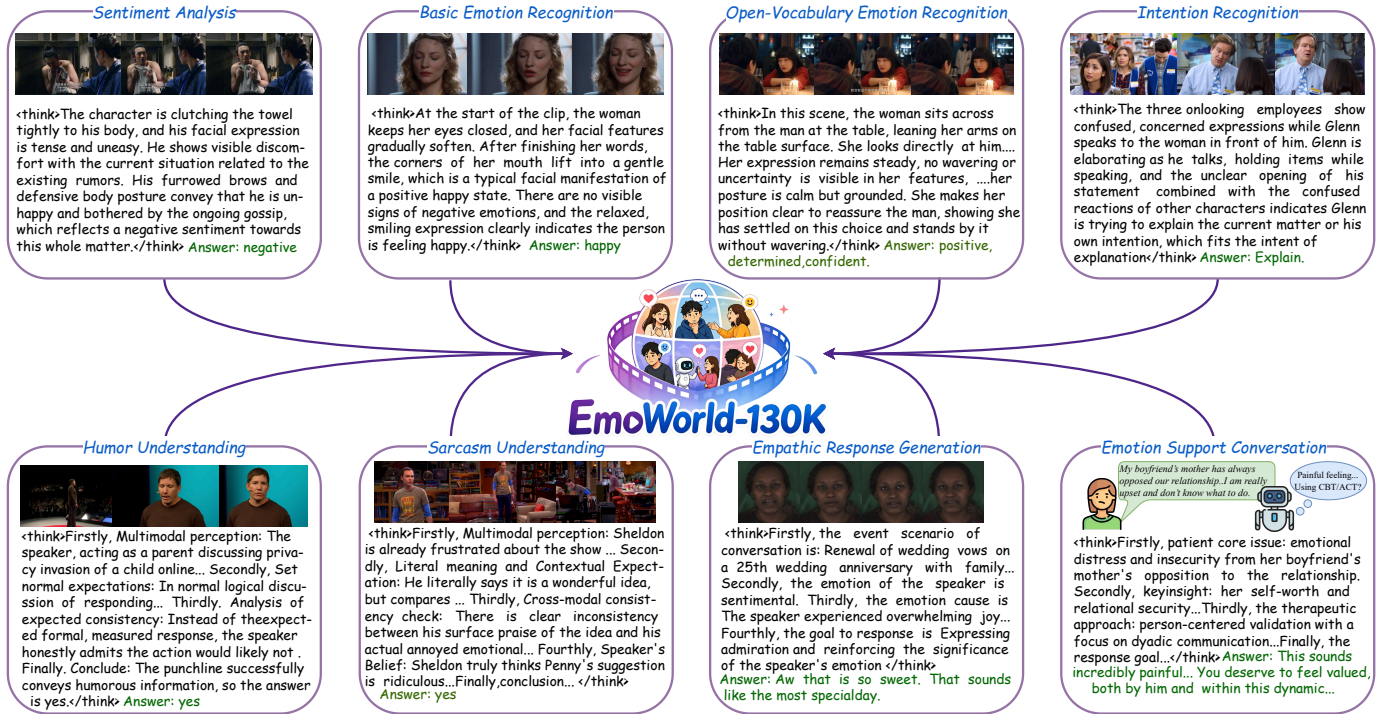


Fig. 1. **EmoWorld-130K Dataset.** This dataset provides high-quality reasoning trajectories for eight affective tasks encompassing three domains: emotion perception (sentiment analysis, basic emotion recognition, and fine-grained emotion recognition), emotion understanding (intent recognition, humor detection, and sarcasm detection), and emotion interaction (empathetic response generation and emotional support conversation).

a hybrid optimization phase combining Group Relative Policy Optimization (GRPO) [17] with a dynamically weighted SFT auxiliary loss. This design enables continuous replay of expert data, preventing capability degradation during exploration. Guided by a synergistic rubric-based reward system, EmoChord enforces reasoning coherence and multimodal factual grounding. Extensive experiments show that OneEmo achieves state-of-the-art results against similarly sized baselines across most benchmarks. Our main contributions are as follows:

- We present **OneEmo**, a unified multimodal reasoning framework capable of handling eight critical affective tasks, spanning foundational emotion perception to advanced multi-turn emotion interactions.
- To build OneEmo, we first introduce **EmoWorld-130K**, a comprehensive dataset featuring psychology-informed reasoning trajectories. Meanwhile, we propose **EmoChord**, a unified multi-task RL strategy that integrates on-policy exploration with off-policy imitation. By using fine-grained reasoning rewards, EmoChord ensures high-quality inference while stabilizing cross-task synergy.
- Extensive automatic and human evaluations demonstrate that OneEmo achieves highly competitive benchmark results and superior interaction capabilities.

II. RELATED WORKS

A. Multimodal Emotion Language Models (MELMs)

MELMs typically comprise a multimodal encoder, an adapter, and a large language model. Benefiting from superior perceptual and reasoning capabilities, MELMs have demonstrated distinct advantages over traditional architectures. Re-

cent advances have leveraged supervised fine-tuning to enable explainable emotion recognition [4], [5], [18]–[20]. Building on this, Zhou et al. [21] integrate structured reasoning with a feedback-driven self-evolution module to achieve fine-grained intent comprehension. Schmidmaier et al. [22] facilitate intent recognition and multi-turn interaction via hierarchical semantic modeling and adaptive reasoning. Lin et al. [7] propose a training-free paradigm for empathetic response generation utilizing prompt engineering and empathetic memory retrieval. Despite these successes, current approaches predominantly target task-specific optimization. This focus overlooks the inherent synergies across diverse affective tasks, even though emotion perception, understanding, and interaction are intrinsically interconnected [15]. To bridge this gap, we introduce OneEmo, a unified multimodal reasoning model designed to comprehensively assimilate emotional domain knowledge.

B. Reinforcement Learning in MELMs

RL has emerged as a pivotal paradigm for eliciting the reasoning capabilities of MELMs. Pioneering works such as R1-Omni [9] and AffectGPT-R1 [13] leverage RL to enhance emotion perception. Subsequent studies, including ERV [23], introduce rationale verifiers to mitigate reasoning-response inconsistencies. Furthermore, EmpRL [24] and MultiMood [14] refine generative quality by utilizing empathy, credibility, and semantic similarity as reward signals. Despite these advances, extending RL to jointly optimize emotion perception, understanding, and interaction remains challenging. The integration of such heterogeneous tasks often leads to unstable optimization dynamics and conflicting gradients. To address this, we

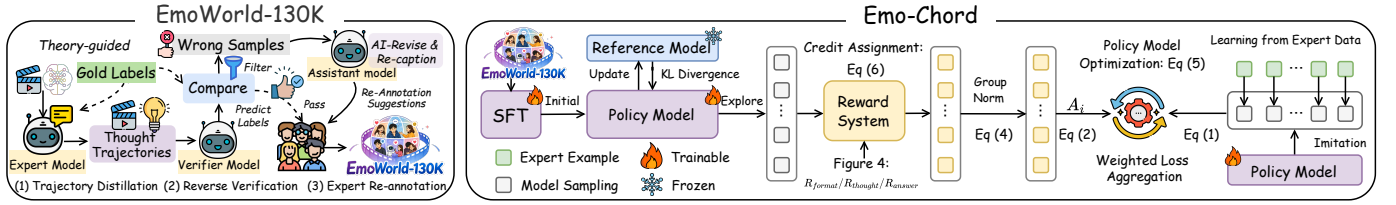


Fig. 2. **OneEmo** utilizes EmoWorld-130K as the training corpus and adopts the Emo-Chord training strategy. **EmoWorld-130K (Left)** is constructed via a three-stage pipeline: reasoning trajectory distillation from expert models, automated answer verification via reverse reasoning, and rigorous manual checks. **Emo-Chord (Right)** initiates with SFT, followed by RL optimization augmented with an auxiliary SFT loss, eliciting the multi-task reasoning capabilities.

introduce the EmoWorld-130K dataset and the Emo-Chord optimization strategy. Together, they enhance the performance and interpretability of MELMs.

III. PRELIMINARY

Current post-training paradigms for MLLMs primarily comprise SFT and RL. Specifically, SFT utilizes curated, domain-specific instruction–response pairs. By learning to mimic the target response distribution, the model minimizes the token-level negative log-likelihood, which is formulated as:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(x,y) \sim \mathcal{D}} \left[\sum_{t=1}^{|y|} \log \pi_{\theta}(y_t \mid x, y_{<t}) \right], \quad (1)$$

where x denotes the multimodal input context and y represents the target token sequence.

RL optimizes a policy model using feedback signals from a reward-based credit assignment system. This paper adopts GRPO [17], which estimates advantages by normalizing rewards across G sampled outputs for each input:

$$\mathcal{L}_{\text{GRPO}}(\theta) = -\mathbb{E}_{(x, \{y_i\}_{i=1}^G) \sim \mathcal{D}} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \min \left(r_{i,t} \hat{A}_i, \text{clip}(r_{i,t}, 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) \right] + \beta \mathbb{D}_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}). \quad (2)$$

Here, a KL divergence penalty regularizes the updated policy π_{θ} toward the reference policy π_{ref} , mitigating policy drift during optimization. The probability ratio $r_{i,t}$ is computed via importance sampling between π_{θ} and π_{ref} :

$$r_{i,t} = \frac{\pi_{\theta}(y_{i,t} \mid x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t} \mid x, y_{i,<t})}. \quad (3)$$

The intra-group relative advantage $\hat{A}_i$ is derived by normalizing the raw reward R_i against the mean and standard deviation of the reward group:

$$\hat{A}_i = \frac{R_i - \text{mean}(\{R_j\}_{j=1}^G)}{\text{std}(\{R_j\}_{j=1}^G)}. \quad (4)$$

IV. METHODOLOGY

Fig. 2 presents the overall pipeline of OneEmo, which takes EmoWorld-130K as the training corpus and Emo-Chord as the optimization strategy. This section elaborates on the dataset construction process and the proposed training pipeline.

A. EmoWorld-130K: Theory-driven Reasoning Dataset

Conventional emotion datasets, e.g., ECR-Chain [25], Psyche-R1 [26], and Emo-CFG [16], predominantly target isolated affective dimensions. This fragmented paradigm neglects the synergistic relationships across tasks and fails to provide comprehensive interpretability. Moreover, these datasets leave the latent reasoning potential of MLLMs underexplored. To bridge this gap, we introduce *EmoWorld-130K*, which is constructed through a three-step workflow, as depicted in Fig. 2 (Left). Statistical distribution and detailed composition of the dataset are summarized in Fig. 3.

1) *Theory-based Trajectory Distillation*: We first source raw data from established datasets, including DFEW [27], MERR [4], MER-Caption+ [5], MER2025-OV [28], MIntRec 1.0/2.0 [29], [30], Mustard [31], URFunny [32], AvaMERG [6], and OpenR1-Psy [33]. Using original annotations as label anchors, we prompt Seed-2.0-Lite [34] to generate structured reasoning trajectories. *Rather than allowing unconstrained generation, we operationalize well-established psychological and cognitive frameworks into explicit prompt templates to rigorously govern the model's reasoning logic.* Specifically, for basic emotion perception, the model extracts multimodal cues and deduces labels based on Ekman's Theory [35], which compels the LLM to anchor its inferences on observable facial expressions and behavioral markers rather than abstract guesses. For humor and sarcasm understanding, it analyzes cognitive expectation mismatches and deciphers pretended attitudes, grounded in the Incongruity-Resolution Theory [36], the Psychology of Humour [37], and the Theory of Irony [38]. For emotion interaction, driven by the Appraisal Theory of Emotion [39], we formulate a four-step reasoning paradigm: dialog memory tracking, user state perception, causal attribution, and response goal formulation. Notably, the emotional support conversations task additionally integrates DSM/ICD-11 diagnostic criteria [40], [41] to standardize clinical state evaluation, and therapeutic strategies (e.g., CBT [42], ACT [43]) to logically dictate the selection of supportive techniques, ensuring highly explainable responses. Compared to the unstructured and verbose thought in OpenR1-Psy [33], our generated trajectories remain highly structured, token-efficient, and strictly context-aware.

2) *Automated Reverse Verification*: To ensure factual consistency and logical coherence in thought annotations, we introduce a closed-loop validation protocol. This protocol assesses whether the gold labels can be deduced solely from

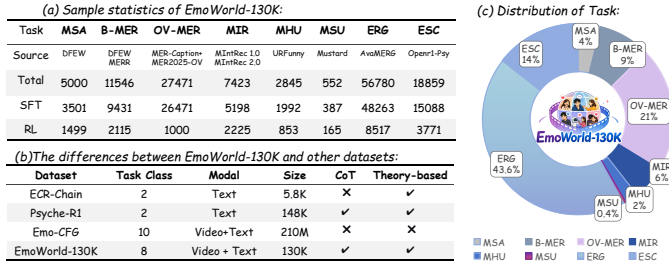


Fig. 3. Statistics of EmoWorld-130K and comparison with other datasets.

the distilled reasoning trajectories. Samples exhibiting inconsistencies are flagged with specific violation types and causal rationales, which are then prioritized for human annotation.

3) *AI-assisted Human Expert Re-annotation*: To address the identified inconsistencies, three psychology postgraduates meticulously reviewed and revised the problematic trajectories, incorporating targeted repair suggestions generated by AI. To establish rigorous quality control, we manually audited a random 10% subset of the data for each task. If any subset failed to meet our stringent quality criteria, we initiated iterative re-annotation and repeated sampling inspections until the entire dataset satisfied our high-quality standards.

B. Emo-Chord: Multi-task Collaborative Optimization

Emo-Chord comprises two core modules: training strategy and credit assignment. Additionally, we introduce a Task-Aware Linear Decay Gating mechanism to regulate reasoning length and mitigate potential hallucinations.

1) *Training Strategy*: To validate that synergistic learning across multi-level affective tasks optimizes overall performance, we first employ a two-stage curriculum learning on the EmoWorld-130K SFT subset (Fig. 3). In the first stage, the model undergoes joint fine-tuning on diverse multimodal tasks to acquire emotion perception and understanding capabilities. In the second stage, we introduce the ESC task and substantially increase the proportion of ERG data. This compels the model to analyze multi-turn dialogues, historical emotional trajectories, and interaction dynamics to generate supportive responses, thereby bridging perception with interaction.

While RL is adept at eliciting reasoning capabilities, we observe that models are susceptible to policy collapse and reward instability, creating significant bottlenecks for RL optimization. Drawing inspiration from recent advances [44], we mitigate this by integrating an auxiliary SFT loss during training:

$$\mathcal{L}(\theta) = (1 - \mu)\mathcal{L}_{\text{GRPO}}(\theta) + \mu\mathcal{L}_{\text{SFT}}(\theta), \quad (5)$$

where μ denotes a dynamic balancing coefficient that controls the relative contribution of each loss component.

Fig. 2 (right) illustrates the overall training pipeline. To circumvent sub-optimal policy entrapment while preserving exploratory diversity, we initiate training with an offline cold-start phase, followed by online policy optimization on *EmoWorld-130K*. This hybrid paradigm effectively cultivates robust multi-task reasoning capabilities while substantially enhancing both training stability and final model performance.

We systematically compare this strategy against alternative pipelines in Section VI-C.

2) *Credit Assignment*: Fig. 4 details the reward functions used in Emo-Chord. For emotion perception, understanding, and interaction, these functions comprise three key components: format, thought, and answer. They are aggregated via coefficients γ_f , γ_t , and γ_a :

$$R = \gamma_f R_{\text{format}} + \gamma_t R_{\text{thought}} + \gamma_a R_{\text{answer}}. \quad (6)$$

R_{format} applies across all tasks as a binary reward enforcing structural compliance: the reasoning process must be strictly encapsulated within `<think>` and `</think>` tags, immediately followed by the final answer:

$$R_{\text{format}}(y) = \begin{cases} 1, & \text{if } y \in \mathcal{F} \\ 0, & \text{otherwise} \end{cases}. \quad (7)$$

R_{thought} and R_{answer} are task-dependent, evaluating the quality of the reasoning trajectory and the final response, to ensure both interpretability and accuracy. We tailor these rewards across two distinct task categories: (a) *Perception and Understanding*: The model infers specific emotional states, dialogue styles, and latent intents from multimodal cues. (b) *Interaction*: The objective shifts to modeling user states, performing causal analysis, and generating empathetic or supportive responses.

Perception & Understanding. R_{thought} comprises factual consistency R_{fact} and reasoning coherence $R_{\text{coherence}}$:

$$R_{\text{thought}} = \frac{R_{\text{fact}} + R_{\text{coherence}}}{10}. \quad (8)$$

To evaluate these components, we devise detailed rubrics for an LLM judge. To compute R_{fact} , a video understanding model first processes the raw video to extract atomic visual fact descriptions. The LLM then compares the visual descriptions generated in the sampled thought process against these objective facts, assigning a discrete score from 1 to 5. This rigorously prevents factual hallucinations and fabricated details from compromising response accuracy. For $R_{\text{coherence}}$, the LLM evaluates whether the semantic trajectory of the reasoning logically aligns with the final answer, ensuring the thought process deduces the outcome naturally without logical leaps.

For R_{answer} , we apply a binary reward for tasks with closed-set label spaces:

$$R_{\text{answer}}(y, o_i^a) = \begin{cases} 1, & y = o_i^a \\ 0, & y \neq o_i^a \end{cases}. \quad (9)$$

For OV-MER, following established protocols [5], [13], we compute the reward based on Emotion Wheel (EW) metrics:

$$F_c(y, o_i^a) = 2 \cdot \frac{\text{Precision}_c(y, o_i^a) \cdot \text{Recall}_c(y, o_i^a)}{\text{Precision}_c(y, o_i^a) + \text{Recall}_c(y, o_i^a)}. \quad (10)$$

$$R_{\text{answer}}(y, o_i^a) = \frac{1}{C} \sum_{c=1}^C F_c(y, o_i^a), \quad (11)$$

where C denotes the number of emotion wheels.

Interaction. R_{thought} comprises a user state modeling reward R_{state} and a strategy alignment reward R_{strategy} . Accurately capturing the user's situational context, emotional state, and

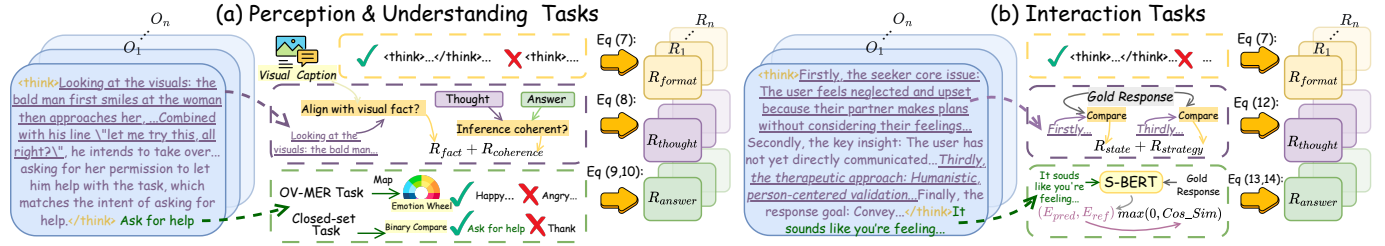


Fig. 4. **Credit assignment in Emo-Chord.** Emo-Chord decomposes rewards into format, thought, and answer components across three domains (perception, understanding, and interaction). For perception and understanding, we enforce constraints on visual details and logical consistency. For interaction, we align the model’s inferred user states and response strategies with reference gold responses, refining the final reward based on semantic similarity.

underlying motivations is critical for response generation. R_{state} evaluates the fidelity of user modeling (scored 1–5 by an LLM judge) by comparing the inferred state against the gold reference. Similarly, R_{strategy} assesses the alignment of the proposed intervention strategy with the gold reference.

$$R_{\text{thought}} = \frac{R_{\text{state}} + R_{\text{strategy}}}{10}. \quad (12)$$

For R_{answer} in these open-ended tasks, we embed the predicted response and the gold response using Sentence-BERT (S-BERT) [45] and calculate their similarity:

$$E_{\text{pred}} = \text{S-BERT}(o_i^a), \quad E_{\text{ref}} = \text{S-BERT}(y_{gt}^a), \quad (13)$$

$$R_{\text{answer}} = \max(0, \text{Similarity}(E_{\text{pred}}, E_{\text{ref}})). \quad (14)$$

3) *Task-Aware Linear Decay Gating*: Emotion tasks exhibit heterogeneous complexity, necessitating variable chain-of-thought lengths. Empirically, we observe that unconstrained models over-generalize verbose reasoning and answers to simple tasks, inducing task-irrelevant hallucinations and degrading performance. To mitigate this, we introduce a task-aware linear decay factor τ to regularize the answer reward R_{answer} . Specifically, we define task-specific soft and hard length bounds. For a generated sequence of length l_o , τ is defined as:

$$\tau = \begin{cases} 1 & l_o \leq l_{\text{soft}} \\ \frac{l_{\text{hard}} - l_o}{l_{\text{hard}} - l_{\text{soft}}} & l_{\text{soft}} < l_o < l_{\text{hard}} \\ 0 & l_o \geq l_{\text{hard}} \end{cases}. \quad (15)$$

We compute decay factors for the reasoning trajectory and the final answer. R_{answer} is then modulated by the minimum of these factors to enforce brevity across both components:

$$R_{\text{answer}} = R_{\text{answer}} \times \min(\tau_{\text{thought}}, \tau_{\text{answer}}). \quad (16)$$

This mechanism adaptively calibrates the answer granularity to specific task requirements, effectively suppressing unbounded verbosity and inefficient reasoning. The statistical procedure for determining these bounds is detailed in the Appendix.

V. EXPERIMENTAL SETUP AND DATASET

A. Implementation Details

OneEmo is trained on EmoWorld-130K using the Emo-Chord strategy using two NVIDIA H800 GPUs. All training phases employ the AdamW optimizer with a micro-batch size of 1 and 8 gradient accumulation steps. The off-policy cold-start phase spans 5 epochs with a learning rate of 1×10^{-5} .

TABLE I

EVALUATION BENCHMARK. TO PREVENT DATA LEAKAGE, WE ENSURE STRICT PARTITION PROTOCOLS. FOR PERCEPTION, TRAINING AND TESTING SAMPLES ARE SOURCED FROM DISJOINT DATASETS; FOR UNDERSTANDING AND INTERACTION, WE ADHERE TO THE OFFICIAL DATASET SPLITS, EMPLOYING THE TRAINING AND VALIDATION SETS FOR TRAINING AND THE TEST SPLITS FOR FINAL EVALUATION.

Task	EmoWorld-130K (Training)	Testing
<i>Emotion Perception</i>		
MSA	DFEW [27]	CMU-MOSI/MOSEI [46], [47], CH-SIMS v1/v2 [48], [49]
B-MER	DFEW [27], MERR [4]	MER2023/2024 [50], [51], MELD [52], IEMOCAP [53]
OV-MER	MER-Caption+ [5], MER2025-OV [28]	OV-MERD [54]
<i>Emotion Understanding</i>		
MSU	MUSTARD [31] (Train/Val)	MUSTARD [31] (Test)
MHU	URFunny [32] (Train/Val)	URFunny [32] (Test)
MIR	MIntRec1.0/2.0 [29], [30] (Train/Val)	MIntRec1.0/2.0 [29], [30] (Test)
<i>Emotion Interaction</i>		
ERG	AvaMERG [6] (Train/Val)	AvaMERG [6] (Test)
ESC	OpenR1-Psy [33] (Train/Val)	OpenR1-Psy [33] (Test)

This is followed by a 2-epoch RL phase, during which the auxiliary loss balancing coefficient μ is annealed from 0.5 to 0.02. For GRPO, we set the learning rate to 2×10^{-6} , the sampling group size to 8, and the KL divergence coefficient to 0.06. The reward coefficients are set as follows: $\gamma_f = 0.5, \gamma_t = 0.3, \gamma_a = 1.0$. In our main experiments, we adopt Qwen3.5-4B [57] as the backbone, unfreezing the multimodal adapter and fine-tuning the language backbone via LoRA. Additional results utilizing alternative backbones are presented in Section VI-E.

B. Benchmarks

Table I summarizes the evaluation benchmarks. For emotion perception, we adopt a cross-corpus evaluation protocol, ensuring that training and testing data are sourced from disjoint datasets. For emotion understanding and interaction, we adhere to the official data splits. The training and validation sets constitute EmoWorld-130K, while the test set is held out for final performance evaluation.

C. Evaluation Metrics

We adopt the official evaluation metrics for each task to ensure fair comparisons. For perception and understanding, we report the Weighted Average F1 (WAF) for MSA [46], Hit Rate for B-MER [5], Emotion Wheel-based metric (EW) for OV-MER [54], and WAF for MIR, MSU, and MHU [29]. For interactions, we employ both automatic and human evaluations. 1) Automatic metrics utilize a 1–5 Likert scale [64], [65], scored by diverse LLMs (GPT-4.1-mini, MiMo-v2.5-pro,

TABLE II

MAIN RESULTS ON EMOTION PERCEPTION AND UNDERSTANDING. THIS TABLE COMPARES ONEEMO AGAINST THREE CATEGORIES OF MODELS: OPEN-SOURCE SPECIALIST MODELS, OPEN-SOURCE GENERALIST MODELS, AND COMMERCIAL GENERALIST MODELS. $\uparrow$ INDICATES THAT HIGHER VALUES ARE BETTER FOR THIS METRIC. THE BEST PERFORMANCE IS SHOWN IN **BOLD**, AND THE SECOND-BEST IS UNDERLINED. ONEEMO OUTPERFORMS OTHER OPEN-SOURCE GENERALIST MODELS AND SPECIALIST MODELS ON MOST TASKS. DESPITE HAVING A SMALLER MODEL SIZE THAN CLOSED-SOURCE MODELS, ONEEMO STILL ACHIEVES COMPETITIVE RESULTS.

Models	Size	MSA					B-MER				OV-MER	MHU	MSU	MIR			Avg.		
		MOSI	MOSEI	SIMS	SIMS-v2	Avg.	MER'23	MER'24	MELD	IEMOCAP	Avg.	OVMERD	UR-FUNNY	MUSTARD	MIRec	MIRec2		Avg.	
		WAF ↑					Hit Rate ↑					EW ↑	WAF ↑	WAF ↑	WAF ↑				
Open-source Specialist Models																			
R1-Omni [9]	2.1B	55.56	48.62	74.71	76.67	63.89	58.30	69.41	40.87	50.18	54.69	51.84	-	-	-	-	-	-	
Nano-EmoX [12]	2.2B	76.82	79.81	86.25	84.76	<u>81.91</u>	79.09	77.94	56.55	60.12	68.43	64.75	-	-	58.17	47.27	52.72	-	
AffectGPT [5]	8.3B	81.30	80.90	88.49	86.18	84.22	78.54	78.80	55.65	60.54	<u>68.38</u>	<u>60.41</u>	-	-	-	-	-	-	
Open-source Generalist Models																			
VidEmo-3B [16]	3.8B	66.37	46.89	76.67	69.28	64.80	40.05	50.37	33.75	29.96	38.53	55.88	48.97	48.93	15.33	5.94	10.64	44.63	
VidEmo-7B [16]	7.7B	65.19	59.46	78.32	76.44	69.85	46.87	55.70	32.95	32.81	42.08	57.98	45.99	53.37	11.57	10.36	10.97	46.71	
MiniCPM-V-2.6 [55]	8.0B	74.96	57.44	74.85	75.04	70.57	46.67	45.31	40.27	36.31	42.14	50.04	61.71	46.63	32.78	23.98	28.38	49.91	
InternVL3.5-4B [56]	4.7B	75.56	72.72	75.05	77.07	75.10	64.23	61.38	52.73	54.00	58.09	56.64	46.00	35.11	49.13	29.87	39.50	51.74	
Qwen3.5-4B [57]	4.5B	74.68	63.25	78.54	79.61	74.02	51.97	55.68	32.44	37.51	44.40	61.52	58.43	55.92	44.94	26.11	35.53	54.97	
Qwen3.5-9B [57]	9.7B	68.60	64.42	80.17	77.97	72.79	52.64	64.51	41.26	43.97	50.60	60.10	54.25	53.28	47.64	33.60	40.62	55.27	
Qwen2.5-VL-7B [58]	7.7B	74.10	58.35	78.65	77.43	72.13	59.81	69.14	48.05	50.53	56.89	55.61	53.28	54.66	53.28	33.54	43.41	56.00	
Qwen3-VL-4B [59]	4.5B	74.02	61.77	79.60	78.20	73.40	52.54	65.36	41.59	47.93	51.86	61.51	71.48	55.90	48.05	29.19	38.62	58.80	
Cosmos3-Nano [60]	16.0B	73.17	76.88	78.24	79.03	76.83	68.71	67.34	50.62	51.72	59.60	58.70	57.28	66.24	58.63	36.99	47.81	61.08	
OneEmo(+Curriculum)	4.5B	76.21	76.71	83.23	82.23	<u>79.60</u>	70.51	73.12	53.05	58.69	<u>63.84</u>	<u>65.40</u>	68.80	<u>69.22</u>	70.48	50.86	<u>60.67</u>	<u>67.92</u>	
OneEmo(+Emo-Chord)	4.5B	75.88	78.13	85.54	84.98	81.13	76.03	80.21	55.06	63.15	68.61	68.54	<u>70.42</u>	74.16	72.48	56.46	64.47	71.22	
Commercial Generalist Models																			
GPT-5-mini [61]	-	80.64	63.17	77.25	77.88	74.74	60.49	58.43	53.36	46.04	54.58	58.80	78.47	62.62	65.21	43.21	<u>54.21</u>	63.90	
Gemini-3.1-Pro [62]	-	76.59	54.98	84.07	84.89	<u>75.13</u>	74.64	78.17	62.72	70.85	71.60	72.90	85.46	<u>70.61</u>	68.31	54.47	61.39	72.85	
MiMo-v2.5 [63]	310B	83.75	66.11	88.05	87.58	81.37	73.21	81.32	52.24	61.77	<u>67.13</u>	<u>65.22</u>	87.04	74.70	62.20	43.96	53.08	<u>71.42</u>	

DeepSeek-v4-flash) to mitigate potential LLM-as-a-judge bias. Specifically, ERG is evaluated on Empathy (Emp.), Contextual Coherence (Coh.), and Informativeness (Inf.), while ESC is assessed via Empathy (Emp.), Skill Use (Skill), and Overall Effect (Ove.). Scores are aggregated using arithmetic means, accompanied by Randolph’s κ for inter-rater reliability. 2) For human evaluation, we conduct blind pairwise comparisons on 100 randomly sampled instances, rated by three psychology undergraduates. This validates Empathy (Emp.) and Humanoid Alignment (Hum.) for ERG, and Overall Effect (Ove.) and Humanoid Alignment (Hum.) for ESC.

VI. RESULTS AND DISCUSSION

A. Main Results

Emotion Perception. Table II presents the zero-shot evaluation results for MSA, B-MER, and OV-MER. OneEmo surpasses all open-source generalists, including 16B-scale multimodal generalists, e.g., Cosmos3-Nano. In comparisons with commercial models, it outperforms models such as GPT-5-Mini and MiMo-v2.5 on B-MER and OV-MER, trailing by only 0.24 points on MSA despite a 70 $\times$ parameter gap. Furthermore, OneEmo optimized via curriculum SFT exhibits highly competitive capabilities in mapping multimodal representational cues into emotional spaces.

Emotion Understanding. As shown in Table II, OneEmo demonstrates exceptional proficiency in conversational expression and social intent recognition. Notably, on complex 20- and 30-class intent recognition, it outperforms Gemini-3.1-Pro by a notable margin of 3.08 points. Operating at an efficient 4.5B parameter scale, OneEmo comprehensively eclipses both open-source affective specialists and generalist models across

perception and understanding levels. Moreover, it remains highly competitive against top-tier commercial models, trailing Gemini-3.1-Pro by a mere 1.63 points. Unlike existing affective specialists, OneEmo goes beyond mere emotion perception; it seamlessly integrates multimodal cue extraction with the context-aware comprehension of human expressions and social intents. These robust outcomes are fundamentally attributed to the Emo-Chord training paradigm. By synergizing environmental exploration with expert imitation, this strategy empowers compact models to overcome parameter constraints, facilitating robust emotional reasoning capabilities.

Emotion Interaction. Table III outlines the LLM evaluation results for ERG and ESC across three dimensions. OneEmo outperforms all open-source baselines, achieving superior empathy in empathetic dialogues and a higher overall effect in emotional support scenarios. Following RL optimization, we observe a marginal decline in informativeness and contextual coherence. We attribute this variance to the human-annotated reference responses in EmoWorld-130K, which intrinsically lack the exhaustive, multi-solution enumerative formats commonly generated by commercial chatbots. However, showing detailed suggestions and proficient psychological techniques is not always appropriate in all situations, and it might even have a negative effect. Expectedly, top-tier commercial models retain an advantage in interaction-level tasks, benefiting from their massive parameter scales and extensive training corpora. Overall, OneEmo delivers highly competitive emotional interaction capabilities within a lightweight parameter footprint.

Table IV presents the results of the blind human evaluation. Across both ERG and ESC tasks, human annotators consistently highlight OneEmo’s exceptional capability to deliver highly anthropomorphic interactions, achieving a remarkable

TABLE III
AUTOMATIC EVALUATION RESULTS ON EMOTION INTERACTION.

Models	ERG			ESC			κ
	Emp. $\uparrow$	Coh. $\uparrow$	Inf.	Emp. $\uparrow$	Skill	Ove. $\uparrow$	
Commercial Generalist Models							
GPT-5-mini	4.40	4.94	4.61	4.27	4.71	4.33	0.45
Gemini-3.1-Pro	4.48	4.97	3.80	4.69	4.58	4.65	0.69
MiMo-v2.5	4.56	4.86	3.78	4.56	4.50	4.52	0.57
Open-source Generalist Models							
Qwen3.5-9B	3.86	4.01	3.16	4.62	4.58	<u>4.51</u>	0.45
Qwen3-VL-4B	<u>4.11</u>	4.07	3.07	4.06	3.11	3.30	0.51
Qwen3.5-4B	3.96	4.22	3.00	4.40	4.21	4.20	0.46
OneEmo(+Curriculum)	3.10	4.38	<u>3.21</u>	4.34	4.07	4.24	0.50
OneEmo(+Emo-Chord)	4.25	4.16	3.10	<u>4.51</u>	<u>4.42</u>	4.67	0.54

TABLE IV
HUMAN EVALUATION RESULTS ON EMOTION INTERACTION.

OneEmo vs.	Dim	ERG			Dim	ESC		
		Win	Tie	Lose		Win	Tie	Lose
GPT-5-mini	Emp.	32.00	19.00	49.00	Ove.	57.00	12.00	31.00
	Hum.	72.00	17.33	10.67	Hum.	60.67	18.67	20.67
MiMo-v2.5	Emp.	41.67	7.33	51.00	Ove.	32.67	34.67	32.67
	Hum.	70.33	10.67	19.00	Hum.	29.00	38.00	33.00
Qwen3.5-9B	Emp.	50.67	3.67	45.67	Ove.	71.33	3.67	25.00
	Hum.	80.33	4.67	15.00	Hum.	84.33	6.67	9.00
Qwen3-VL-4B	Emp.	49.67	4.00	46.33	Ove.	45.33	37.67	17.00
	Hum.	78.33	7.00	14.67	Hum.	50.33	39.00	10.67

The differences in win/loss counts were statistically significant across all dimensions (p -value < 0.01), except for a statistical tie against MiMo-v2.5.

TABLE V
ABLATION STUDY OF REWARD FUNCTIONS.

R_{format}	R_{thought}	R_{answer}	MSA	B-MER	OV-MER	MIR	MSU	MHU	ERG	ESC
×	×	×	74.02	44.40	61.52	35.53	58.43	55.92	3.73	4.27
×	×	✓	80.25	65.60	63.64	57.40	68.02	73.77	3.75	4.03
✓	×	×	80.36	63.68	66.05	59.68	61.57	72.87	3.80	3.97
✓	×	✓	80.79	68.18	66.80	64.32	70.02	68.41	3.62	4.25
×	✓	✓	80.81	65.16	65.46	59.52	65.81	70.44	3.76	4.12
✓	✓	✓	81.13	68.61	68.54	64.47	70.42	74.16	3.84	4.53

peak win rate of 84.33% in human-likeness. Nevertheless, OneEmo trails GPT-5-mini and MiMo-v2.5 on ERG task, it still exhibits highly competitive performance against massive proprietary models. In the complex ESC task, it decisively surpasses GPT-5-mini and achieves a statistical tie with the 310B-parameter MiMo-v2.5. Furthermore, when evaluated against state-of-the-art open-source baselines, our model establishes comprehensive dominance in both interaction scenarios. These subjective assessments compellingly validate that our proposed framework translates robustly into superior human-centric emotional engagements in the real-world.

B. Impact of Reward Functions

As shown in Table VI-A, the full reward configuration yields comprehensive performance enhancements. Specifically, ablating the answer reward leads to a substantial average decline of 2.97 points across eight tasks. Similarly, removing the thought and format rewards results in average decrements of 1.16 and 2.58 points, respectively. While answer and format rewards directly guarantee outcome accuracy and structural compliance, the thought reward uniquely anchors multi-level

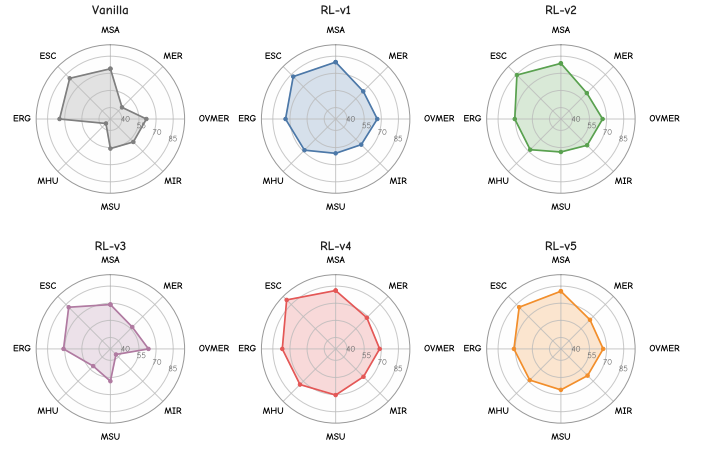


Fig. 5. **Training strategy comparison.** We linearly scaled the evaluation metrics for ERG and ESC by a factor of 20 to align their numerical ranges with those of other tasks.

capabilities. Specifically, it enforces multimodal factual consistency and reasoning coherence in perception tasks, while supervising user state modeling and strategy selection in interactive scenarios. By imposing these fine-grained semantic constraints, our design prevents reliance on spurious correlations. Consequently, final predictions are firmly grounded in logically sound reasoning processes, ultimately driving OneEmo’s exceptional multi-task generalization.

C. Impact of Different Training Strategy

We conduct comprehensive studies to validate the design choices of our training strategy. Table VI summarizes the training details and variants. Fig. 5 presents the performance of diverse RL paradigms compared to the vanilla baseline. Directly applying joint RL and SFT optimization without a prior off-policy cold-start (RL-v3) yields severely degraded performance, particularly struggling on complex tasks like MIR and MHU. Without foundational multimodal grounding, the model fails to secure consistent reward signals across heterogeneous tasks, leading to unstable optimization. Furthermore, regarding the optimal transition point for RL, we observe that initializing exploration from the first stage of Curriculum SFT (RL-v1, RL-v4) consistently outperforms initialization from the fully completed second stage (RL-v2, RL-v5). We attribute this to policy entrenchment: exhaustive supervised curriculum training inadvertently over-solidifies sub-optimal behaviors, restricting the model’s exploratory capacity during subsequent RL phases. Among all paradigms, our proposed *Emo-Chord* (RL-v4) achieves the optimal balance and highest aggregate gains. By coupling a strategically calibrated, limited cold-start with dynamic multi-task exploration synergized with expert data replay, *Emo-Chord* effectively circumvents both reward sparsity and policy entrenchment, converging to vastly superior reasoning policies.

D. Task Synergy Analysis

Fig 6 illustrates the model performance under varying SFT data compositions. By examining the incremental integration

TABLE VI
DIFFERENT MODEL VARIANTS AND THEIR TRAINING DETAILS.

Variants	Training Pipeline	Training Details
Vanilla	—	Raw backbone
SFT-v1	Vanilla+SFT(P)	SFT on perception tasks
SFT-v2	Vanilla+SFT(P+U)	SFT on perception and understanding tasks
SFT-v3	Vanilla+SFT(P+U+I)	SFT on perception, understanding, and interaction tasks
SFT-v4	Vanilla+CoT-free SFT(P+U+I)	SFT-v3 without explicit thinking supervision
SFT-v5	Vanilla+Curriculum SFT(P+U+I)	SFT-v3 with two-stage curriculum learning
RL-v1	Vanilla+Curriculum SFT(S1)+RL	Initialized from the first stage of Curriculum SFT and optimized via GRPO
RL-v2	Vanilla+Curriculum SFT(S2)+RL	Initialized from the second stage of Curriculum SFT and optimized via GRPO
RL-v3	Vanilla+CoT-free SFT	Direct GRPO training with auxiliary SFT loss
RL-v4	Vanilla+Curriculum SFT(S1)+RL mix SFT (Emo-Chord)	Initialized from the first stage of Curriculum SFT and optimized via RL mix SFT
RL-v5	Vanilla+Curriculum SFT(S2)+RL mix SFT	Initialized from the second stage of Curriculum SFT and optimized via RL mix SFT

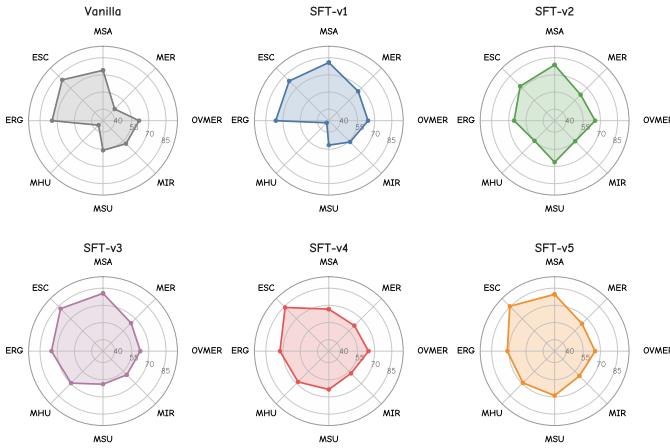


Fig. 6. **Task synergy analysis.** We conduct an ablation study to quantify the impact of varying training data compositions in SFT.

of tasks from basic perception to empathetic generation (SFT-v1 to v3), we observe that naive joint training inherently suffers from task interference. While it enhances the actively trained tasks, it inadvertently degrades the performance of untrained domains, failing to foster effective cross-task synergy. Furthermore, the explicit thought-augmented variant (SFT-v3) maintains a consistent overall advantage over the direct-reasoning baseline (SFT-v4), suggesting the added value of incorporating explicit thought trajectories. Ultimately, by employing strategic data mixing and staged curriculum learning, SFT-v5 successfully navigates these bottlenecks. It demonstrates robust inter-task mutual promotion, effectively circumvents catastrophic forgetting, and yields a highly balanced comprehensive performance.

E. Generalization Across Different Backbones

To evaluate the model-agnostic efficacy of our dataset, we apply LoRA fine-tuning to two distinct vision-language foundations: Qwen3-VL-4B and InternVL-3.5-4B. As illustrated by the pre- and post-training comparisons in Fig. 7, both architectures exhibit strict Pareto improvements across the evaluated affective tasks. This compellingly demonstrates the robust cross-model generalizability of EmoWorld-130K. Regardless of the underlying architecture, our dataset systematically elevates the models' competencies from foundational

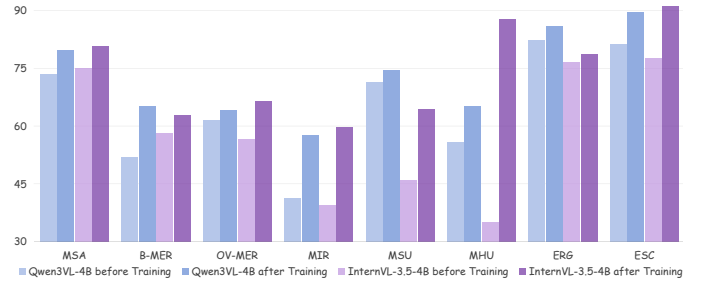


Fig. 7. **Performance across different backbones.** This figure compares the pre-training and post-training performance of two backbones: Qwen3-VL-4B and InternVL-3.5-4B. Experimental results demonstrate that our methods yields consistent improvements across all eight tasks, confirming the strong generalization capability of our approach.

perception to deep comprehension, ultimately unlocking highly reliable empathetic interaction capabilities while successfully avoiding inter-task capability conflicts.

VII. CONCLUSION

This paper introduces **OneEmo**, a unified reasoning model designed explicitly for emotional intelligence. To facilitate this, we present **EmoWorld-130K**, a comprehensive dataset featuring explicit reasoning trajectories across eight typical emotion tasks, and **Emo-Chord**, a novel multi-task reinforcement learning strategy that integrates online-offline collaborative policy optimization with well-designed reward allocation. Extensive experiments demonstrate that OneEmo consistently surpasses same-scale baselines on most benchmarks, validating the efficacy of explicit reasoning and inter-task synergy. Furthermore, comprehensive ablation studies verify the contribution of each module and elucidate the impact of distinct reward functions. This work marks an important step toward advancing unified artificial emotional intelligence.

VIII. ETHICS STATEMENT

EmoWorld-130K is derived exclusively from publicly licensed datasets without involving new human-subject data collection, and all human annotations were conducted by fairly compensated volunteers under strict anonymity and informed consent. Furthermore, although our emotional support tasks leverage clinical frameworks as reasoning scaffolding, OneEmo remains strictly a research prototype and must not

substitute for professional psychiatric diagnosis or treatment. We explicitly advise against its unsupervised clinical deployment. Consequently, any real-world application necessitates the integration of strict crisis-referral guardrails, and both the dataset and model will be released under a restrictive license rigorously enforcing these ethical and safety boundaries.

IX. LIMITATIONS AND FUTURE WORKS

While our work represents a substantial stride toward unified artificial emotional intelligence, certain limitations remain. First, emotion is inherently context-sensitive and culturally diverse in its behavioral manifestations. Current explainable datasets often rely on scripted cinematic data; thus, future benchmarks must prioritize authentic, in-the-wild interactions. Second, although our framework effectively bridges perception, understanding, and interaction, real-world deployments demand an even broader task spectrum, such as continuous emotion forecasting and long-term empathetic companionship. Future research will focus on collecting diverse real-world data, fusing physiological and behavioral signals, and integrating these advanced capabilities to construct more robust, generalized affective agents.

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I. APPENDIX OVERVIEW

In this Appendix, we present the following as an extension to the ones shown in the paper:

- Additional Details (section II)
- Prompting Protocols (section III)
- Qualitative Analysis (section IV)

II. ADDITIONAL DETAILS

A. Distribution of Data Length

As illustrated in Fig. 1, the natural output lengths of EmoWorld-130K vary significantly across different task paradigms. For instance, perception or understanding tasks (e.g., B-MER, MIR tasks) typically yield concise answers. In contrast, interaction-level tasks (e.g., OV-MER and ESC task) intrinsically necessitate more extensive reasoning trajectories and detailed conversational responses. To accommodate these heterogeneous length biases, we heuristically assign dynamic thresholds grouped by the inherent length characteristics of each task category. This task-specific thresholding method achieves a vital pragmatic balance: it effectively suppresses pathological verbosity, such as reward hacking or endless rambling during RL exploration, without truncating valid, well-formed reasoning chains. Crucially, as corroborated by Fig. 1, the vast majority of our samples naturally fall within the penalty-free region. This geometric alignment indicates that our heuristic thresholds successfully preserve the natural distribution of the model’s output, strictly penalizing only genuinely excessively long generations.

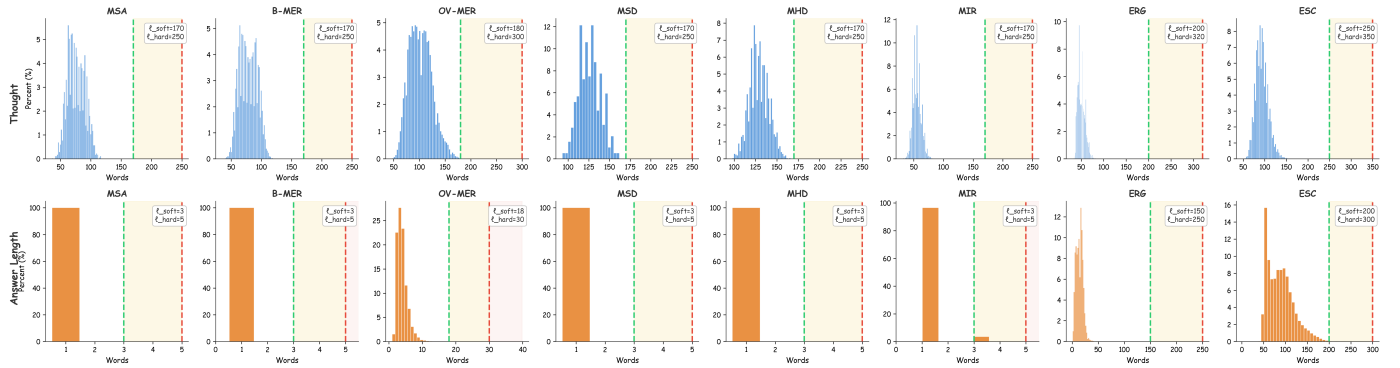


Fig. 1. **Length distributions of reasoning trajectories and responses in EmoWorld-130K.** The green and red dashed lines denote the task-specific soft and hard length penalty thresholds, respectively. Notably, the vast majority of samples naturally fall within the penalty-free region, demonstrating that our heuristic thresholds preserve the natural output distribution while exclusively penalizing pathological verbosity.

B. Training Data Ratio

TABLE I
DATA MIXING PROPORTIONS (%) ACROSS DIFFERENT TRAINING PHASES.

Phase	MSA	MER	OVMER	MIR	MSU	MHU	ERG	ESC
Curriculum Stage 1	7.26	9.69	58.11	10.90	0.73	3.63	9.69	-
Curriculum Stage 2	3.64	5.46	45.47	8.55	0.69	3.46	18.19	14.55
Emo-CHord	9.43	11.31	16.50	15.32	1.29	6.58	28.28	11.31

Table I details the task sampling proportions across different training phases. Stage 1 prioritizes fundamental perception for robust multimodal grounding, while Stage 2 dynamically amplifies interaction tasks. Finally, *RL mix SFT* introduces ESC and balances the multi-level distribution to foster comprehensive affective synergy without catastrophic forgetting.

III. PROMPTING PROTOCOLS FOR AUTOMATED EVALUATION AND RUBRIC SCORING

In this section, we provide the complete prompt templates in the file `prompts.py`, which can be found at our github code repository. Below we present a condensed version of the key instructions:

A. Perception and Understanding Task Thought Reward Prompt

Instructions:

Use only the provided information. Ignore fluency and formatting unless they affect judgment. Inferring internal states from visual cues is permitted as long as they do not contradict the provided facts.

Task: Evaluate the candidate’s reasoning on multimodal tasks (e.g., emotion recognition, intent detection) by distinguishing objective atomic visual behaviors from subjective inferences. Assessment should be based on the extracted behaviors and reasoning content.

1. Visual Fact Consistency

Assess alignment of the model’s internal reasoning with provided atomic visual facts. Penalize fabricated observations, invented events, or contradictions. Allow reasonable high-level inferences if not contradicted by observable cues.

Scoring Criteria:

5 points: No claims clearly contradict provided facts or explicit text; no substantial hallucinations.

4 points: Essentially consistent. Minor unsupported details present but do not affect core judgment.

3 points: Partially consistent. Some important inferences are weakly grounded, but overall logic remains acceptable.

2 points: Multiple significant claims lack basis, are exaggerated, or involve selective interpretation of evidence.

1 point: Most reasoning relies on fictional cues or clearly conflicts with evidence.

0 points: No usable basis, missing or irrelevant reasoning, or reasoning explicitly contradicted by evidence.

2. Reasoning-Answer Coherence

Assess internal consistency between reasoning and the final answer (evaluate coherence, not answer correctness). The answer must naturally follow from the reasoning (convergence for single-label; logical support for open-label).

Scoring Criteria:

5 points: Reasoning clearly and directly supports the final answer, with no contradictions or unresolved alternatives.

4 points: Essentially consistent. Answer matches reasoning, with only minor ambiguity or brevity.

3 points: Roughly compatible, but the logical chain from reasoning to answer is incomplete, vague, or omits key clarifying steps.

2 points: Clear mismatch. Some reasoning points in a different direction, or the final answer lacks sufficient support.

1 point: Severe contradiction between reasoning and final answer.

0 points: Missing reasoning or answer, or answer cannot be logically inferred from reasoning.

B. Interaction Task Thought Reward Prompt

Instruction: Evaluate the reasoning process for empathetic/counseling tasks (ERG/ESC) by comparing candidate reasoning with reference reasoning. Focus on semantic alignment, not wording style.

1. User State Alignment

Criteria: Compare at a semantic level. Focus on capturing the user’s situation, emotional state, core conflict, needs, and drivers. Exact wording is not required; equivalent paraphrases are accepted.

Scoring Criteria:

5: Fully aligned. Captures the user’s state, emotions, needs, and core conflict with no material drift.

4: Mostly aligned. Minor omissions or wording differences, but the core state is captured.

3: Partially aligned. Some central aspects are captured, but important emotional or situational details are missing or blurred.

2: Weak alignment. Captures only a small part of the user state, or mixes it with notable misunderstanding.

1: Largely misaligned. Describes a substantially different user state or emotional meaning.

0: No usable user-state analysis, or irrelevant to the reference reasoning.

2. Response Strategy Alignment

Criteria: Compare the implied strategy with the reference. Focus on the intended response plan, sequencing, and prioritization (e.g., validation, exploration, reassurance, reframing).

Scoring Criteria:

- 5: Fully aligned. Matches the reference strategy, support goal, and prioritization of moves.
- 4: Mostly aligned. Same broad strategy with only minor omissions or reduced specificity.
- 3: Partially aligned. Relevant but misses an important component of the reference strategy.
- 2: Weak alignment. Only superficial overlap or a noticeably off-target direction.
- 1: Strong mismatch. Substantially different or counterproductive strategy.
- 0: No usable strategy, or irrelevant to the reference reasoning.

General Instructions:

C. Interaction Task Evaluation Prompt

##ERG task:

Instruction: Evaluate the response on a scale of 1 (Poor) to 5 (Excellent) across the following dimensions.

Dimension 1: Empathy

Score Rubric:

- 1: Completely misses or contradicts the seeker's emotional state; response is cold, dismissive, or overtly inappropriate.
- 2: Shows minimal recognition of emotion; response is generic (e.g., "I'm sorry to hear that") with no evidence of true understanding.
- 3: Acknowledges the emotion at a surface level; response is adequate but lacks depth, specificity, or genuine warmth.
- 4: Demonstrates clear understanding of the emotional state; response is warm, validating, and reasonably tailored to the seeker's situation.
- 5: Deeply and precisely captures the nuanced emotional state; response feels genuinely human, highly validating, and emotionally resonant.

Dimension 2: Coherence & Consistency

Score Rubric:

- 1: Completely incoherent, contradictory, or irrelevant to the seeker's message.
- 2: Mostly inconsistent or contains noticeable logical gaps; relevance is weak.
- 3: Generally coherent and relevant, but contains minor inconsistencies, vague connections, or slight drift.
- 4: Consistent and well-connected to the context; logical flow is clear with only trivial issues.
- 5: Perfectly coherent, logically airtight, and seamlessly aligned with the seeker's context and intent.

Dimension 3: Informativeness

Score Rubric:

- 1: Purely templated or meaningless filler; adds zero informational or conversational value.
- 2: Mostly generic platitudes with negligible substantive content; could apply to any situation.
- 3: Contains some meaningful content but still relies noticeably on generic or templated language.
- 4: Substantive and reasonably specific; provides genuine value (insight, relevant question, useful framing) with minimal templating.
- 5: Highly informative and uniquely tailored; offers profound insight, a precisely targeted question, or genuinely useful perspective with no detectable templating.

##ESC task:

Instruction: Evaluate the response on a scale of 1 (Poor) to 5 (Excellent) across the following dimensions.

Dimension 1: Empathy

Score Rubric:

- 1: Misses, dismisses, or contradicts the patient's feelings; cold or invalidating.
- 2: Minimal acknowledgment of emotion; generic or perfunctory.
- 3: Recognizes the general feeling but remains surface-level.

- 4: Clearly understands and validates the patient's emotional state with warmth and presence.
 5: Precisely captures the emotional nuance and feels deeply human, attuned, and validating.

Dimension 2: Support Skill

Score Rubric:

- 1: No discernible support strategy, or actively harmful technique (e.g., toxic positivity, blame, excessive advice).
- 2: Attempts a strategy but executes poorly (e.g., closed yes/no question, irrelevant self-disclosure, shallow reassurance).
- 3: Uses one or more recognizable support strategies correctly, but selection or timing is suboptimal (e.g., advice before validation).
- 4: Deploys appropriate strategies with good timing and execution; demonstrates competent conversational support skills.
- 5: Masterful, context-sensitive use of multiple support strategies with impeccable timing; reflects professional-level support skill.

Dimension 3: Overall Effectiveness

Score Rubric:

- 1: Likely to leave the seeker feeling worse, unheard, or frustrated; response is counter-therapeutic.
- 2: Minimal positive impact; seeker would likely feel little change or mild disappointment.
- 3: Moderately helpful; seeker receives some comfort or direction, but the response lacks lasting impact or full trust.
- 4: Clearly helpful; seeker would likely feel noticeably heard, validated, and somewhat emotionally relieved.
- 5: Exceptionally effective; seeker would likely feel deeply understood, genuinely comforted, and measurably better equipped to cope.

IV. QUALITATIVE ANALYSIS



Task: Intent Recognition; GT: *Asking for Opinions*;
 Transcript: "Penny: what are you gonna do?"

Question: [MIR] Watch the video and figure out what the speaker is trying to do. Which intention best matches their behavior and words? [Intent Candidate List].



OneEmo

Thinking: The blonde woman in the red dress asks "what are you gonna do?" while speaking to the man carrying a laundry basket who just entered the bathroom area. She is prompting the man to respond about what he intends to do next, which aligns with the intent of asking for the other person's opinion on his next action. There are no conflicting visual cues, so the social intent is asking for opinions.

Answer: *Asking for opinions*



InternVL-3.5-4B

Thinking: In the video, the blonde woman asks the man "what are you gonna do?", which is a question directed at the man's upcoming action. The man responds that he is going to the store, which answers the question. The woman's inquiry is intended to get information about the man's next action, which matches the intent of asking for information.

Answer: *ask for help*



Cosmos3-Nano

Thinking: Okay, let's try to figure out what the speaker's intention is here. The user provided the video and the question the character asks: "what are you gonna do?" \n\nFirst, I need to recall the context. The woman in the video is in a tense situation, possibly a confrontation or a moment of high emotion. Her expressions shift from serious to slightly amused, and she's in a confined space like a hallway or elevator. The setting includes signs like... it's likely she's using it to provoke a reaction, maybe in a ...

Answer: *taunt*

Fig. 2. Qualitative comparison of reasoning trajectories on a multimodal intent recognition task. While OneEmo accurately grounds both visual and textual cues to deduce the correct intent, the baseline models exhibit label mapping errors (InternVL-3.5-4B) and severe contextual hallucinations (Cosmos3-Nano).

Fig. 2 compares reasoning trajectories for intent recognition. OneEmo accurately extracts multimodal cues (e.g., characters and objects) to logically deduce the correct intent. In contrast, InternVL-3.5-4B correctly interprets the semantic inquiry but fails in label mapping. Meanwhile, Cosmos3-Nano exhibits severe contextual hallucinations, fabricating a tense confrontation in an elevator, resulting in an erroneous taunt prediction.